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ORIGINAL ARTICLE

Objectively Measured Light Exposure in Emmetropic and Myopic Adults

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ABSTRACT

Purpose. Light exposure has a close link with numerous aspects of human physiology including circadian rhythm, mood disorders, metabolism, and eye growth. Here, a lightweight wrist-worn device was employed to continuously measure light exposure and activity across seasons and between refractive error groups to assess objectively measured differences and compare with subjectively reported data.

Methods. Subjects, aged 21–65 years (n = 55), wore an actigraph device (Actiwatch Spectrum) continuously for 14 days to quantify light exposure, activity, and sleep. Subjects were classified as emmetropic (n = 18) or myopic (n = 37), and answered an activity questionnaire. Additionally, devices were calibrated against a lux meter and UV sensor for indoor and outdoor settings, and used to measure ambient illumination in various environmental conditions.

Results. Subjects spent 1:52 ± 0:56 hours outside per day, as measured objectively. Subjectively reported measures overestimated objective measures by 0:25 ± 1:19 hours per day (range −1:49 to +4:29 hours, $P < .05$). Subjects spent 1:04 hours more outdoors in summer and received an increased cumulative light dose compared to winter ($P < .005$). There were no significant differences in objective measurements of time outdoors between myopic and emmetropic subjects. Ambient illumination measures from the Actiwatch correlated with a lux meter for all locations tested ($R^2 = 0.99$, $P < .001$). Ambient illumination was highest in the summer at 176,497 ± 20,310 lux and lowest for indoor artificial light at 142 ± 150 lux.

Conclusions. Subjects spent more time outdoors and received an increased light dose in summer, with no differences between refractive error groups in this adult population. Various environmental and seasonal measurements revealed significantly different available light in winter versus summer and indoors versus outdoors. Objective devices such as the Actiwatch can be valuable in studies where quantification of environmental factors is critical.

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Key Words: light exposure, myopia, refractive error, outdoor time, activity, actigraphy

Light exposure has a close link with numerous aspects of human physiology and has been implicated in several different pathological processes including circadian rhythm disturbances,[1] mood disorders,[2,3] cancer,[4] and metabolic disorders.[5,6] Of relevance to the current study, light exposure may also play a role in eye growth and myopia.[7–9]

The prevalence of myopia has been increasing and is considered epidemic in some populations,[10] with the most recent figures in some groups exceeding 80 to 90%.[11,12] Because of the ocular risks[13] and the high socioeconomic burden associated with high myopia, there is great interest in elucidating the risk factors contributing to myopic eye growth. It is well established that genetics plays a role in the development of myopia in children.[14] However, the rapidly increasing prevalence of myopia indicates that environmental factors, such as near work and light exposure, also play a role in eye growth.[15] Based on studies such as COMET,[16] CLEERE,[17] and others,[18] near work seems to make only a small contribution to the prevalence of myopia.

Several studies have reported an association between time spent outdoors and the prevalence of myopia in children[14,19–22] and young adults.[23] Read et al. found that myopic children exhibited significantly lower daily light exposure than emmetropic children in Australia.[8] A study in young adults found higher UV exposure in stable versus progressing myopes.[24] A meta-analysis of seven cross-sectional studies reported a 2% reduced odds of myopia per additional hour of time spent outdoors per week.[25]

Seasonal differences in light exposure may also play a role in eye growth patterns. Gwiazda et al. found slower progression of myopia in summer and speculated this was likely a result of children spending more time outdoors and less time in school during the summer.[21,26] Another study in Chinese children found

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that myopia progression in the summer months was 60% of that seen in winter.[27]

The majority of previous studies evaluating outdoor activity in children have been done through parent questionnaires.[19,20,22] More recently, wearable light sensors have been utilized.[8,28] Alvarez and Wildsoet used an arm-mounted HOBO pedant light sensor to evaluate light exposure in young adults in three seasons and administered a questionnaire. The authors found that results from the questionnaire were in poor agreement with objective data.

With accumulating evidence to suggest that time outdoors plays a role in eye growth, the current study was designed to evaluate the use of a wearable light sensor and accelerometer to objectively measure light exposure and activity patterns in an adult population. The device has been actively used in both children and adults in various applications, most notably in sleep studies.[29–31] However, the device has not been used widely to measure light exposure. The model utilized here is capable of measuring the spectral composition of light in terms of the red, green, and blue components, making the device ideal for evaluating the effects of wavelength-specific effects.

The purpose of the current study was twofold: to establish the validity of the Actiwatch Spectrum in objective measurements of light exposure and spectral composition in the environment, and to correlate objectively measured seasonal differences of light exposure in adults to subjective questionnaires while assessing potential differences between refractive error groups in a cross-sectional analysis.

METHODS

This study utilized 22 actigraph devices (Actiwatch Spectrum; Philips Respironics, USA) in an environmental calibration study and measurement of light exposure and activity patterns in Houston, Texas, USA. All subjects provided informed consent. The study was approved by the Committee for Protection of Human Subjects at the University of Houston and followed the tenets of the Declaration of Helsinki.

The Actiwatch Spectrum is a wrist-worn actigraph device that measures ambient illumination and activity at 32 Hz. The light sensor consists of color-sensitive photodiodes to measure illuminance of white light in lux (range 0.1–200,000 lux), and the irradiance of the red, green, and blue components in microwatts per square centimeter ($\mu W/cm^2$, wavelength range 400–700 nm). According to the manufacturer, the chromatic channels of the Actiwatch Spectrum are as follows: red, 600–700 nm; green, 500–600 nm; and blue, 400–500 nm. Physical activity is measured via a solid-state piezoelectric accelerometer and is expressed in counts per minute (CPM). Based on a previous study suggesting that light exposure sampling between 10 and 120 seconds is stable,[28] a 30-second epoch was used here. The battery life and memory allow for more than 2 weeks of data collection at this sampling rate.

Device Calibration

Five randomly chosen Actiwatch devices were measured against a calibrated research lux meter (Gossen Starlight, Germany) and a UV sensor (Solar Light, USA) in 14 different conditions, including inside 10 buildings on the University of Houston Campus and 4 outdoor locations. The five devices were mounted on a holder with the light sensors directed upward. For each location, the devices recorded ambient light for 5 minutes, and five individual measurements were recorded with the lux meter and UV sensor.

To evaluate the specificity of the spectral sensors on the Actiwatch Spectrum, five of the devices were tested against a bank of red (625 nm), green (525 nm), and blue (460 nm) LEDs (Eco Series Modules; NC LED, Korea). The LEDs have a bandwidth of 10 nm, and for each color, nine individual LEDs were mounted together. The five devices were directed towards the LED set-up and placed 50 cm from the light source in a dark room. The bank of green LEDs was illuminated on its own for 5 minutes while the devices recorded, followed by the red LEDs, then blue LEDs. Finally, all three sets of LEDs were illuminated together while the devices recorded light levels for five more minutes. The overall quantum count of the LEDs was not controlled for. Therefore, the absolute value of the monochromatic light is not considered here; rather, the relative light of the illuminated channel is compared to the non-illuminated channels.

Environmental Ambient Illumination

Five devices were used to measure light levels in various conditions, including winter sun and shade, summer sun and shade, various rooms in homes, and an elementary school classroom. For each condition, the devices were mounted on a holder with the sensors directed upward for 7 days. For outdoor conditions, ambient illumination measures during the brightest 2 hours of the day were averaged across 7 days and then averaged for all devices. For indoor conditions, one of two techniques was used. For the elementary school classroom, only weekdays were considered, and ambient illumination measures were averaged over school hours, 8:00 AM to 3:00 PM. For locations in homes, data were averaged between 7:00 PM and 9:00 PM to characterize typical artificial indoor lighting. In addition, two devices were tested outdoors with standard sunglasses (14% transmission; Spexan, Humphrey Instruments) placed over the device sensors at 10 mm and directed upward. The devices were tested with sunglasses for 10 minutes in full sun and 10 minutes in cloud cover during the summer.

Light Exposure and Activity Patterns in Adults

Fifty-five subjects participated in this study. Subjects were classified as emmetropic (n = 18) or myopic (n = 37) based on history and habitual correction. Subjects completed a custom-written activity questionnaire, including estimates, in hours per week, of various activities such as time spent at work, working on a computer, reading books, and outdoors. Specifically, subjects were asked to estimate how many hours they spent outdoors during daylight on a typical weekday (Monday–Friday) and a typical weekend day (Saturday–Sunday). The data were then weighted to calculate the average time spent outdoors per day using the following equation:

$$\text{Time outdoors per day} = ((\text{weekday time} \times 5) + (\text{weekend day time} \times 2))/7 \quad (1)$$

Subjects then wore an Actiwatch Spectrum device continuously for a 2-week period between January and November 2014. Subjects were instructed not to remove the device, including during sleep. The device contains an "off-wrist" sensor to monitor compliance. Subjects were asked to ensure that the light sensor was unobstructed

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TABLE 1.
Values used for activity and ambient light grading

CPM	Activity Grading
<319	Inactive
320–1,047	Light activity
1,048–1,623	Moderate activity
>1,624	Vigorous activity
Light Level (lux)	Light Grading
<9	Darkness
10–99	Dim indoor light
100–999	Standard indoor light
1,000–9,999	Standard outdoor light
>10,000	Bright outdoor light

CPM, counts per minute.

and not covered by clothing. Light, activity, and sleep data were grouped based on the month and day the device was worn. Seasons for 2014 were distributed as follows, based on the solstices and equinoxes for Houston, Texas: winter, January 1–March 19 (n = 15); spring, March 20–June 20 (n = 19); summer, June 21–September 21 (n = 15); and fall, September 22–December 20 (n = 6). The number of daylight hours for each season was calculated based on sunrise and sunset times for the latitude and longitude location of the University using the solar calculator of the Global Monitoring Division of the Earth System Research Laboratory (National Oceanic Atmospheric Administration).

Data Analysis

The Actiwatch software (Actiware 6.0.4; Philips Respironics, USA) was utilized to determine activity and sleep parameters, including bed time, wake time, and duration. Mean daily activity was calculated by averaging counts per minute (CPM) for the hours the subject was awake. Light exposure and activity data were only included in analyses when the device was worn for the entire day. Days were excluded if the subject removed the device for more than 30 minutes, or if the light exposure dropped to zero for 30 minutes or more during daylight hours, indicating that the sensors on the device were obstructed. Nights were excluded if the subject removed the device for all or part of the night.

Results are presented as mean values ± standard deviation. Device calibration data were analyzed with linear regressions. Subjective and objective data were analyzed with paired two-tailed t-tests and a Bland-Altman plot. Light exposure, activity, and sleep data were analyzed across seasons and between refractive error groups with unpaired two-tailed t-tests. Activity and light parameters were adapted from previous validation studies using a similar wrist-worn Actiwatch accelerometer in adults[28] and were also used in a recent publication evaluating light exposure in children (Table 1).[8]

RESULTS

Device Calibration

Ambient illumination measured from the Actiwatch correlated with the lux meter for 14 indoor and outdoor locations tested (R^2 = 0.99, $P < .001$). There was no relationship between values measured with the Actiwatch and those measured with the Solar Light Meter when indoors (due to the lack of UV indoors). However, an increase in UV was correlated with higher ambient illumination outdoors.

The chromatic channels of the Actiwatch showed high specificity for monochromatic light (Fig. 1). When the devices recorded ambient illumination from a bank of nine red LEDs, the red channel recorded 39.49 ± 15.3 μW/cm^2 with the green and blue channels recording 0.03 ± 0.03 μW/cm^2 and 0.01 ± 0.01 μW/cm^2, respectively. When the green LEDs were illuminated, the green channel recorded 40.28 ± 6.66 μW/cm^2, with the green and blue channels recording 0.43 ± 0.08 μW/cm^2 and 5.07 ± 1.09 μW/cm^2, respectively. When the blue LEDs were illuminated, the blue channel recorded 91.24 ± 20.41 μW/cm^2, with the red and green channels recording 0.33 ± 0.07 μW/cm^2 and 4.18 ± 0.82 μW/cm^2, respectively.

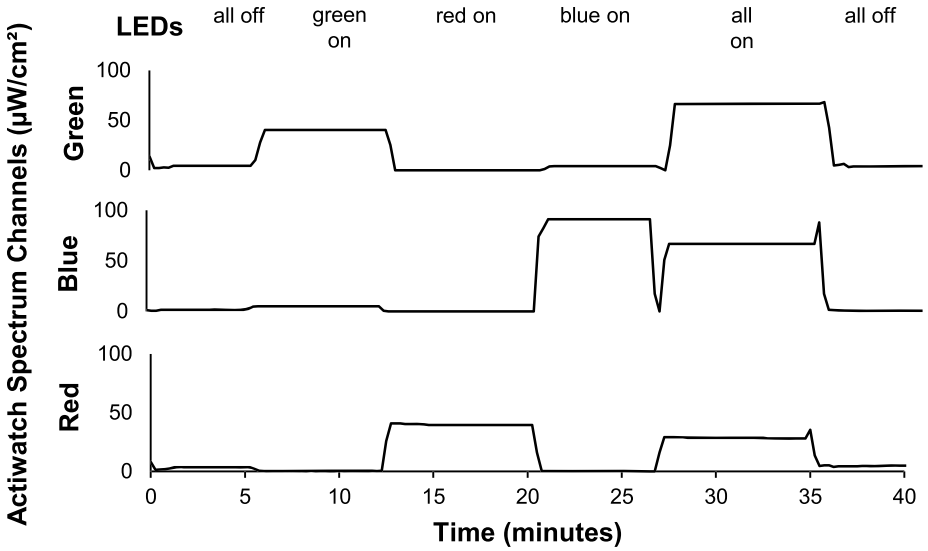


FIGURE 1.
Output of the red, green, and blue chromatic channels of the Actiwatch Spectrum (n = 5) tested against monochromatic red, green, and blue LEDs (measured in μW/cm^2); LEDs were turned on individually and all together over time.

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TABLE 2.
Ambient light (lux) and spectral composition ($\mu W/cm^2$) for various indoor and outdoor conditions

Condition	Ambient light	Red	Green	Blue
Summer sun	176,497 ± 20,310	4,175 ± 1,044	15,127 ± 826	5,655 ± 2,167
With sunglasses	34,207 ± 12,223	3,836 ± 602	2,842 ± 1,206	759 ± 505
With cloud cover and sunglasses	2,973 ± 1,049	270 ± 162	259 ± 88	96 ± 41
Summer shade	2,313 ± 269	163 ± 29	246 ± 6	96 ± 19
Winter sun	80,517 ± 42,535	3,677 ± 1,697	7,376 ± 3,905	3,454 ± 1,986
Winter shade	1,443 ± 622	58 ± 41	139 ± 60	63 ± 31
Classroom	248 ± 168	14 ± 8	24 ± 16	8 ± 6
Home				
Family room	295 ± 5	38 ± 1	23 ± 0.4	8 ± 0.1
Office	3 ± 1	0.3 ± 0.2	0.3 ± 0.06	0.02 ± 0.01
Bedrooms	25 ± 2	8 ± 3	11 ± 7	5 ± 4

Environmental Ambient Illumination

Mean ambient illumination, measured over a 7-day period and averaged between five devices, was brightest during the midday summer sun (Table 2). Ambient illumination greater than 199,999 lux were not measurable by the device. Therefore, time points that recorded an error because the value was too high were later replaced with 200,000 lux, and the mean therefore might represent a small underestimation. The occurrence was rare and was observed only during full summer sun. With this substitution, the summer sun during the brightest two hours of the day was 176,497 ± 20,310 lux. In the summer shade, ambient illumination was 2,313 ± 269 lux. For the two devices that were tested outdoors in summer sun with standard dark sunglasses over the light sensor, ambient illumination was 34,207 ± 12,223 lux in full sun and 2,973 ± 1,049 lux in cloud cover. For full winter sun, ambient illumination was 80,517 ± 42,535 and for winter shade was 1,443 ± 622 lux. For indoor environments, the elementary school classroom had an ambient illumination during school hours (8:00 AM to 3:00 PM) of 248 ± 168 lux. During evening hours, ambient illumination of a family room was 294 ± 5 lux, a home office was 3.15 ± 1 lux, and children's rooms were 25 ± 2 lux. For all indoor conditions, green and red were the highest spectral component and blue was the lowest.

Light Exposure and Activity Patterns in Adults

Adult subjects were 37.0 ± 8.8 years of age (range 21–64 years) and consisted of 24 males and 31 females. The mean number of days and nights included in the analysis from the device were 13.2 ± 1.4 and 14.2 ± 1.3, respectively. Activity and light exposure patterns for a representative subject for weekdays and weekends are shown in Fig. 2.

Subjective Versus Objective Measures

According to the survey, subjects reported an average of 2:12 ± 1:18 hours outdoors (Fig. 3). This was significantly different than the average time outdoors recorded objectively by the device, which was 1:52 ± 0:56 hours ($P < .05$). Bland-Altman analysis showed a mean difference of 0:25 ± 1:19 hours. A total of 58% of subjects overestimated their time outdoors (by up to 4:29 hours), and 42% of subjects underestimated their time outdoors (by up to 1:49 hours).

According to the survey, emmetropic subjects spent 2:48 ± 1:44 hours outdoors per day and myopic subjects spent 1:56 ± 0:59 hours outdoors per day, with a significant difference of 48 minutes per day ($P < .05$). According to the device, emmetropic subjects spent 2:05 ± 1:06 hours outdoors per day and myopic subjects spent 1:46 ± 0:51 hours outdoors per day, with a difference of 19 minutes per day. However, this difference did not reach statistical significance ($P = .26$).

Seasonal Variations

Time spent in various light levels from >1 to >100,000 lux in spring, summer, fall, and winter is shown in Fig. 4. When analyzed across seasons, objectively measured outdoor time was highest for subjects who wore the device in the summer at 2:21 ± 1:00 hours (Fig. 5A), which was significantly higher than winter outdoor time (1:17 ± 0:51 hours, $P < .005$) by 1:04 hours. Spring and fall outdoor time were not significantly different than summer, at 1:59 ± 0:34 hours ($P = .22$) and 1:44 ± 1:10 hours ($P = .24$), respectively. Time spent outdoors in the spring was not significantly different than in the fall ($P = .50$). Calculations from the solar calculator show an average number of daylight hours for winter was 11:03 ± 0:41, for spring was 13:15 ± 0:46, for summer was 13:17 ± 0:32, and for fall was 11:05 ± 0:43. Depending on the season, subjects were outdoors for 15 ± 2.5% of the available daylight hours.

Mean daily white light exposure was 1,669.46 ± 1,176.09 lux (range 128.7–5,353.07 lux; Fig. 5B). Daily light exposure was highest during the summer at 2,353.23 ± 1,411.0 lux and lowest in the winter at 1,060.89 ± 869.55 lux ($P < .01$). Spring and fall daily light exposure fell in between summer and winter, at 1,608.36 ± 713.93 lux and 1,674.97 ± 1,698.21 lux, respectively. Daily white light exposure for emmetropes was 1,758.54 ± 1,384.85 lux and for myopes was 1,611.55 ± 1,055.4 lux. However, this difference in white light exposure, and for red, green, and blue, did not reach statistical significance ($P = .66$, 0.74, 0.58, and 0.65, respectively; Fig. 6).

Mean daily activity was 298 ± 76 CPM (range 169–560 CPM; Fig. 5C). Activity was not significantly different for any season

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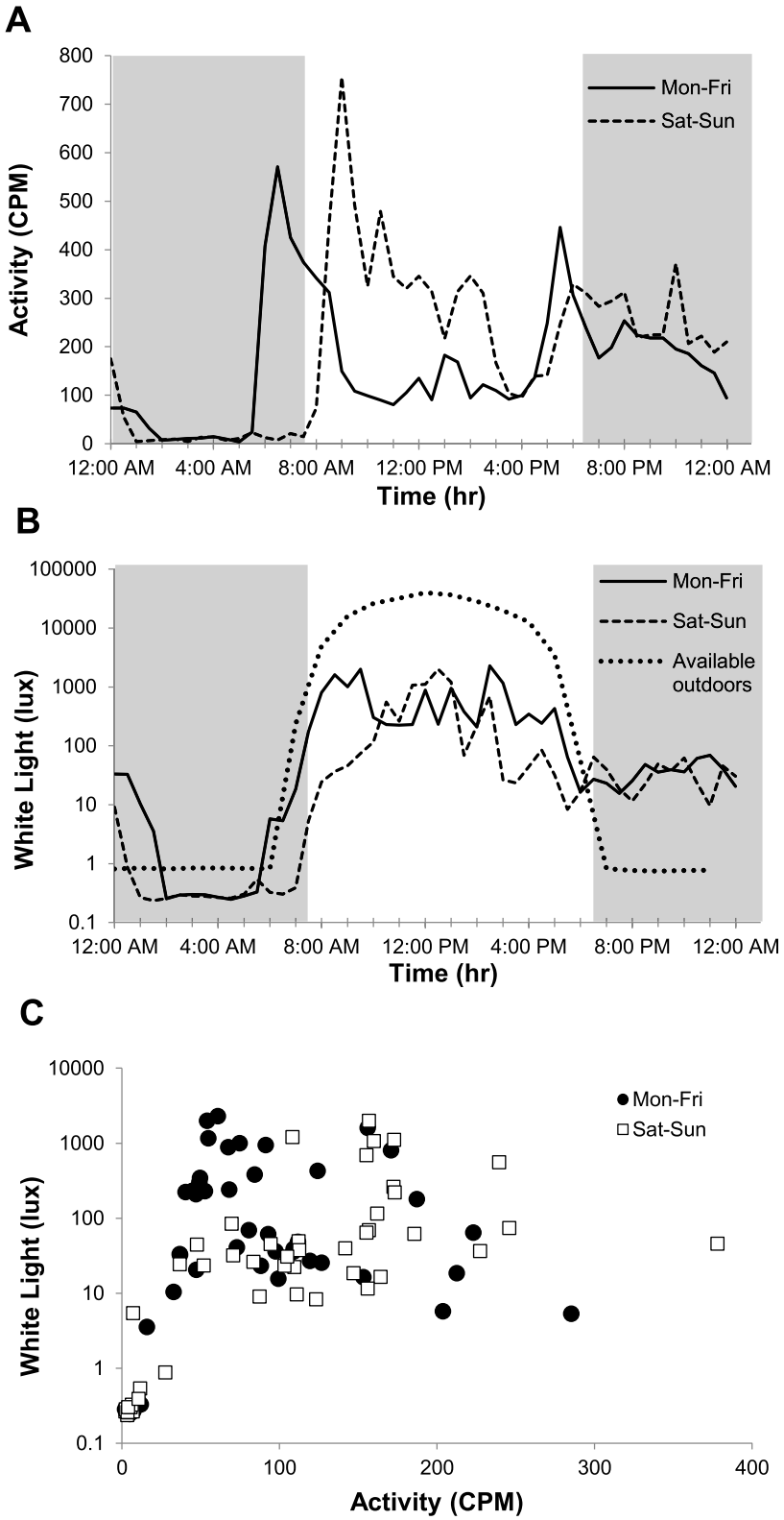


FIGURE 2.

(A) Activity (counts per minute) and (B) light exposure (lux) patterns for a representative subject (age 38 years) on weekdays (Mon–Fri, solid line, n = 10) and weekend days (Sat–Sun, dashed line, n = 4). Gray area represents sundown. Dotted line in (B) represents available light outdoors. (C) Relationship between activity and white light exposure on weekdays (closed symbols) and weekend days (open symbols) for the same subject shown in (A) and (B).

($P > .05$ for each season against all other seasons). There was no difference in daily activity between emmetropes and myopes ($P = .78$). Higher average daily activity was correlated to increased time spent outdoors ($P < .01$; Fig. 7).

Average wake time was 7:01 AM ± 1:02 hours (range 5:23 AM to 10:59 AM) and bed time was 11:17 PM ± 0:54 hours (range 9:39 PM to 1:56 AM). Average sleep duration was 7:41 ± 0:49 hours (range 6:11 to 9:11 hours). Sleep duration was not significantly different

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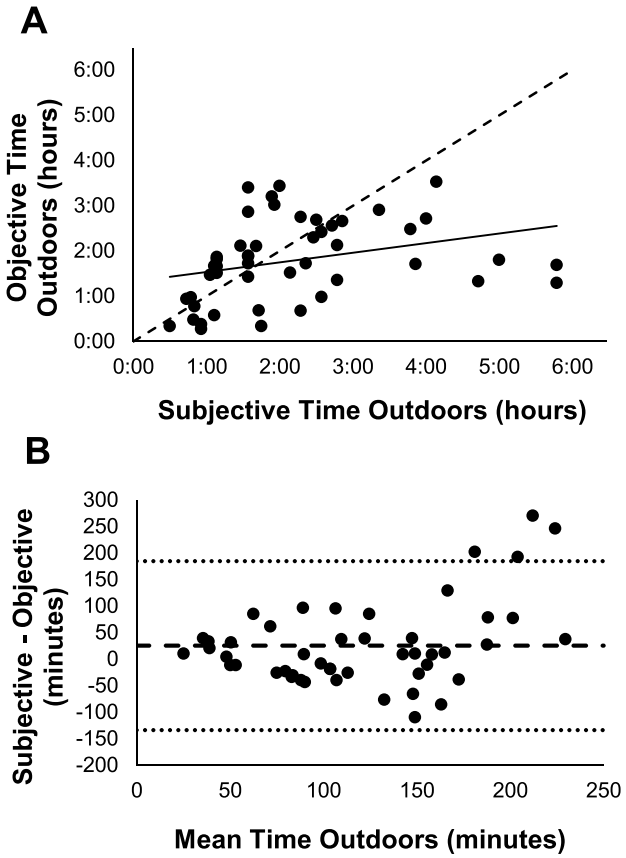


FIGURE 3.

(A) Hours spent outdoors measured subjectively with the survey against hours spent outdoors measured objectively with the Actiwatch spectrum for all subjects (n = 55). Dashed line represents the 1:1 relationship, and the solid line represents the linear regression of the data. (B) Bland-Altman analysis shows a mean difference of 25.56 minutes (dashed line) with a confidence interval of ±159.31 minutes (dotted lines).

between any season ($P > .05$ for each season against all other seasons; Fig. 5D). There was no difference in sleep duration between refractive error groups ($P = .95$).

DISCUSSION

Subjects spent more time outdoors and received an increased cumulative light dose in summer months, whereas activity and sleep were found to be similar across seasons, as measured objectively. There were no differences in light exposure and activity patterns between refractive error groups for this adult population. Various environmental and seasonal conditions revealed significantly different available light between winter and summer months and indoors versus outdoors. Ambient illumination and relative contributions of red, green, and blue were characterized for common indoor and outdoor conditions and across seasons for a subtropical location using the Actiwatch Spectrum. Light exposure in various geographical locations, and differences in lifestyle between ethnic groups, should be considered to fully understand contributions to eye growth. Objective monitoring techniques such as this have potential in determining the contributions of light and wavelength-specific properties in future eye growth and circadian rhythm–related studies.

Subjects spent over an hour more per day outdoors in the summer than in the winter (2:21 vs. 1:17 hours) as measured objectively. Additionally, ambient illumination was more intense in the summer than in the winter. Therefore, an hour spent outdoors in the summer would result in a higher light dose than in the winter. These differences in time spent outdoors and light intensity between summer and winter may help to explain why eye growth in children slows in summer months compared to school sessions.[26]

Based on previous studies,[8,28] a scale was devised to characterize light levels from dim to high illumination. Ambient illumination is exponentially lower indoors than outdoor environments, with

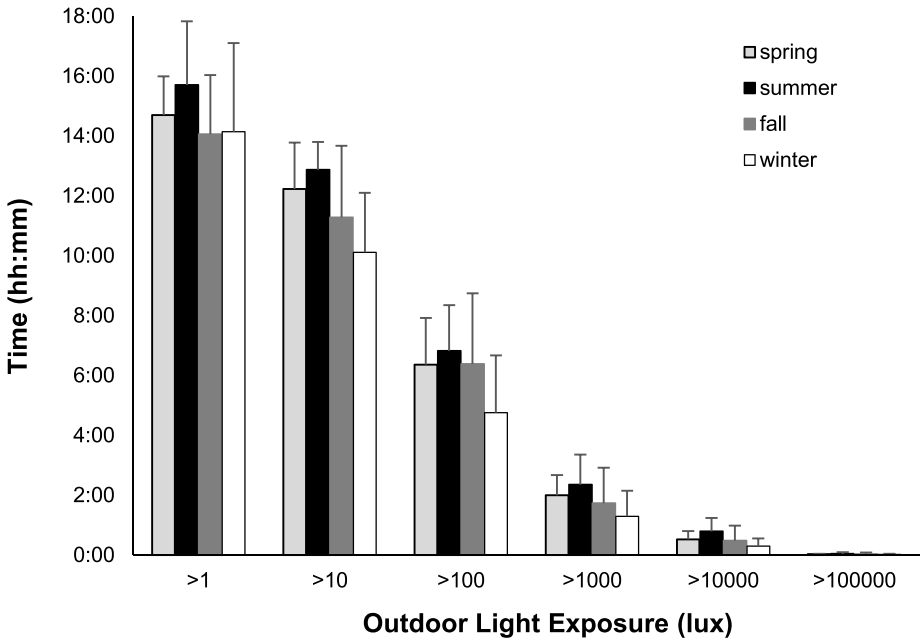


FIGURE 4.

Time spent in various light levels from >1 to >100,000 lux in spring (n = 19, light gray bars), summer (n = 15, black bars), fall (n = 6, dark gray bars), and winter (n = 15, white bars).

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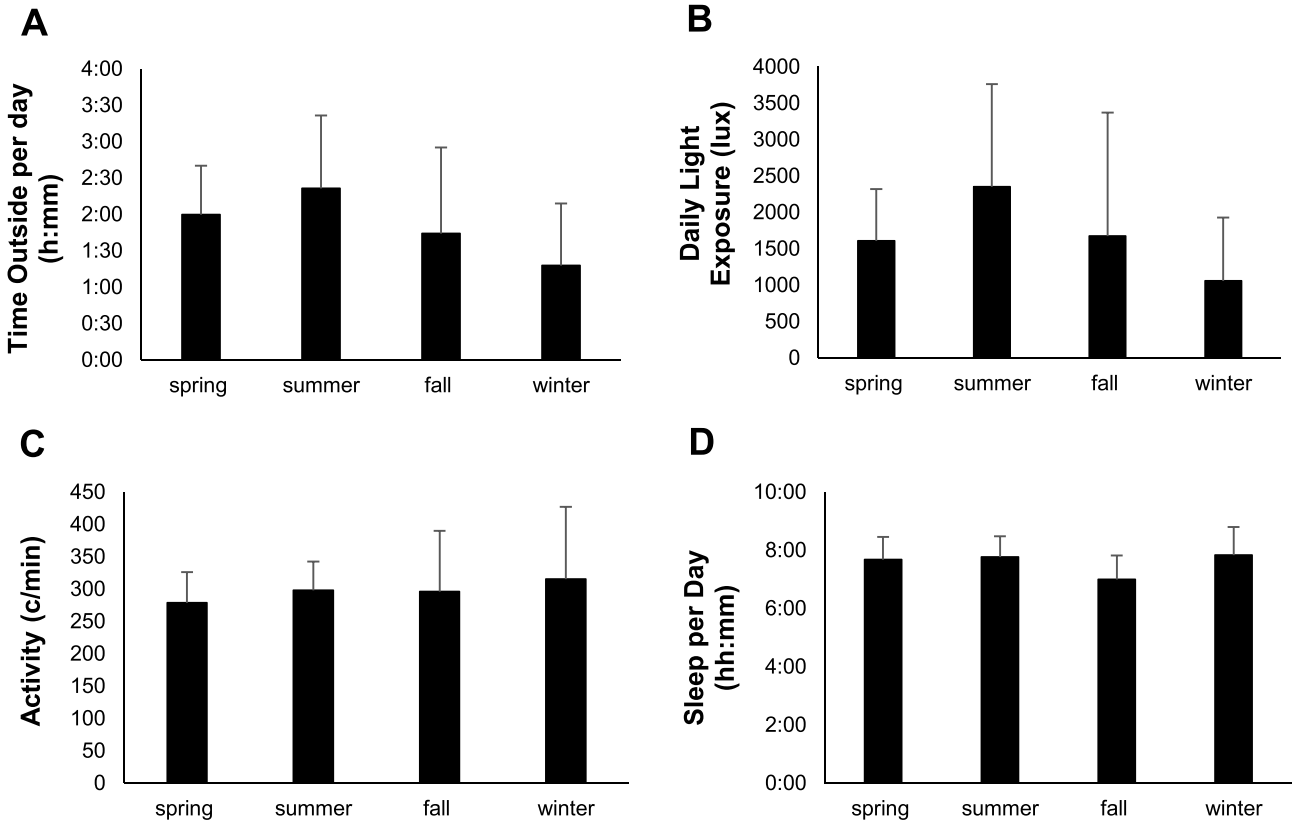


FIGURE 5.

(A) Time spent outdoors (h:mm), (B) average daily white light exposure (lux), (C) average daily activity level (CPM), and (D) sleep duration (h:mm) analyzed across seasons for all subjects (spring n = 19, summer n = 15, fall n = 6, winter n = 15).

no indoor values recorded over 1,000 lux. Therefore, ambient illumination measures over 1,000 lux were assumed to be outdoors. In the summer sun, when people are likely to be wearing sunglasses, the ambient illumination with standard sunglasses placed over the device sensors was recorded to be over 30,000 lux, which is significantly higher than indoor levels. In addition, in full shade in the summer, ambient illumination was over 2,000 lux. Some cultures, such as in Australia, show a low prevalence of myopia and spend increased time outdoors compared to other cultures. Even though children are required to wear hats and sunglasses

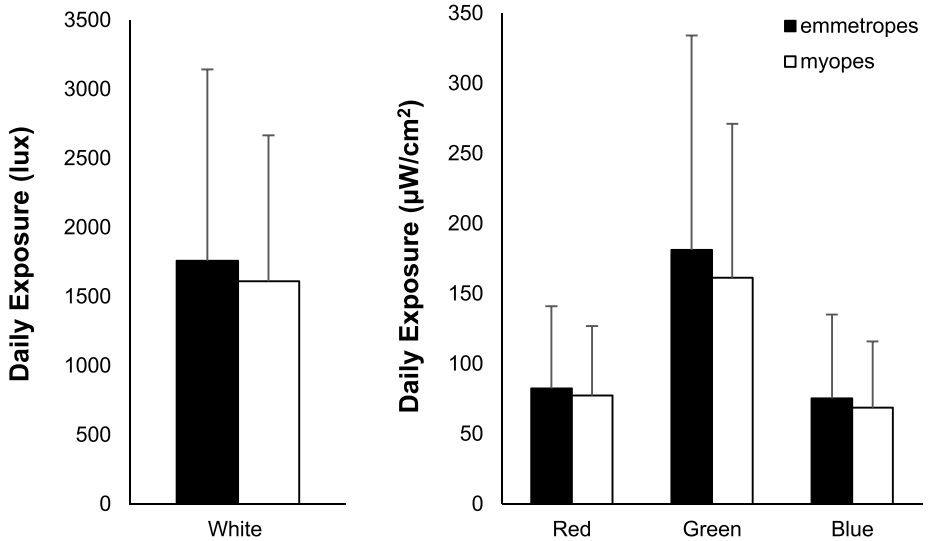


FIGURE 6.

Average daily white light exposure (lux) and the intensity of the red, green, and blue components ($\mu W/cm^2$) in emmetropes (black bars, n = 18) and myopes (white bars, n = 37). Error bars represent standard deviation.

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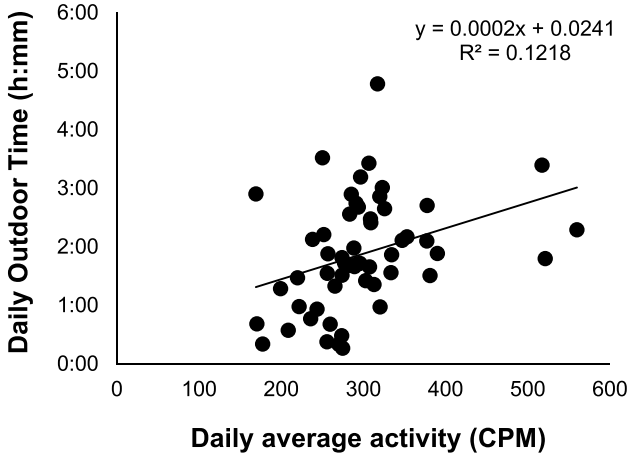


FIGURE 7.

Relationship between daily average activity (CPM) and times spent outdoors (h:mm) for all subjects (n = 55).

outdoors, ambient illumination reaching their eyes is log units above indoor lighting and may contribute to the lower prevalence of myopia.

Objectively measured time outdoors was not significantly different for emmetropic and myopic adults. Other studies have also reported no differences in time spent outdoors in adults between refractive error groups.[28] However, this does not exclude the possibility that children in different refractive error groups may spend different amounts of time outdoors. Subjects here wore the devices in various seasons, which confounds comparisons across refractive error groups. Additionally, refractive error groups were determined by subject report and absolute refractive error was not measured.

Objectively measured time outdoors was significantly different than estimated in the questionnaire, with some subjects overestimating time spent outdoors and others underestimating. One possible source of this discrepancy may be from time spent in the car. The device records high light levels when driving, whereas the subject might not have accounted for this as "time spent outdoors." In addition, questionnaires are subject to poor recall and subject biases. Use of the wearable electronic monitoring devices, such as the Actiwatch, affords the opportunity to objectively and precisely quantify light levels and time spent outdoors, and correlate ambient illumination with activity levels.

Another objective marker of light exposure is conjunctival autofluorescence (UVAF), in which UV-induced fluorescence of the nasal and temporal conjunctiva is photographed and measured. This method has been shown to be reliable and to correlate with the level of outdoor activity in adults as reported in questionnaires.[32] Using this method, authors found that myopic refractive error was inversely associated with conjunctival UVAF in a young adult population.[33] However, disadvantages of this method over light and activity measurements presented here are that conjunctival UVAF measured in cross-sectional populations may represent a cumulative measure of lifetime light exposure. Therefore, specific time periods of high light exposure cannot be distinguished, such as seasonal differences in light exposure and their correlation to activity levels.

Spectral composition of light may play a role in myopia and eye growth; however, evidence have been inconsistent.[34–36] Guinea pigs and chicks raised in monochromatic short-wavelength light became more hyperopic compared to those raised in longer wavelength or broadband light.[35,37] The effects of monochromatic light on refractive error may be caused by longitudinal chromatic aberration, with shorter wavelengths focusing in front of the retina and longer wavelengths behind the retina. A study in rhesus monkeys showed that a small proportion of animals became myopic in red light, whereas the majority remained hyperopic.[36] Another study showed that rhesus monkeys wearing red lenses became relatively hyperopic.[34] These results suggest that mechanisms other than chromatic aberration may be involved, with complex interactions between S, M, and L cone sensitivity. A study in Chinese students showed that students with red-green color vision deficiencies were more likely to be myopic than those without color vision deficiencies.[38]

The intrinsically photosensitive retinal ganglion cells (ipRGCs) located in the inner retina respond directly to short-wavelength blue light and are responsible for non–image-forming functions including circadian rhythm entrainment and pupil size.[39,40] Recent evidence that short-wavelength blue light is protective against myopia development[35] leads to the hypothesis that ipRGCs activity may play a role in eye growth. ipRGCs project onto dopaminergic amacrine cells and may be involved in dopamine modulation with increased light exposure.[9,41] The Actiwatch Spectrum, with chromatic light sensors, and algorithms to determine activity and sleep patterns, may help to quantify the effects of the spectral composition of light on eye growth and circadian rhythm patterns to further test this hypothesis.

The Actiwatch Spectrum presents some limitations. The device must be uncovered by clothing to record accurate ambient light measurements, which may present a problem in colder environments. It is apparent in the data if the watch is obstructed because the light exposure drops to zero despite the presence of activity. Subjects were reminded not to cover the device, and when the data were observed to drop to zero for more than 30 minutes during daylight hours, the day was not included in the analysis. Another limitation is the location of the device on the wrist as opposed to near the eyes. However, the correlation between light exposure measured at eye level versus the wrist has been reported to be 0.76.[42] Hats and sunglasses are not detected by the device, although outdoor measurements when the sensors were covered with sunglasses were over >1,000 lux and would still be considered outdoor levels. Finally, the light sensor was not able to record ambient illumination above 200,000 lux. Only a small number of points for some subjects during the summer reached 200,000 lux. Because any measurement above 1,000 lux is considered to be "outdoors," this ceiling is unlikely to affect the results.

In conclusion, wearable monitors, such as the Actiwatch Spectrum, are ideal for objective and continuous measurement of light exposure, activity, sleep, and other aspects of human behavior. Further studies in which devices are utilized longitudinally in children may help elucidate the role of light in eye growth and clarify the influence of intensity and wavelength. Studies such as these can potentially offer modifiable behaviors to decrease the development and progression of myopia.

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